Appendix

Compliance in regulatory gray areas: The case of the organic seed standard

Survey sample

1. Data collection method

- 2010: Data were collected by convenience sample only, shared by 23 organizations, including non-profits and foundations, organic certifiers, and departments of agriculture. Surveys were distributed primarily over email, but paper, phone, and in-person surveys were conducted at the request of the grower. We received a total of 1,027 responses (712 web-based, 315 paper-based), 539 of which included complete enough data for our analysis.
- 2015: Data were collected by both random and convenience sampling. For the random sample, 800 certified organic producers were randomly selected from the approximately 12,000 organic producers that use seed, recorded in the NOP database. To encourage responsiveness, three follow-up emails were sent following an initial email, paper copies were mailed on request, and individual phone calls were made to follow-up. From these 800, 197 (25%) responded (114 web-based, 83 paper-based) and 122 (16%) were complete enough to use in our analyses. For the convenience sample, surveys were shared by 41 organizations, primarily over email. We received a total of 1,148 responses from the convenience sample, 649 of which included complete enough data for our analysis. In total we had 771 usable responses.
- 2020: Data were collected by both random and convenience sampling using methods similar to 2015. For the random sample, 2,000 certified organic producers were randomly selected from the the NOP database and given the option to complete the survey over email or paper. After an initial email and postcard invitation, growers received follow-up contacts by email, postcard, and phone, and hard copies of the survey were sent. Survey distribution was managed by Washington State University's Social and Economic Sciences Research Center (SESRC) and distributed alongside the Organic Farming Research Foundation's (OFRF) National Organic Research Agenda. From these 2,000 surveys, 448 (22%) responded (206 web-based and 242 paper-based) and 185 (10%) were complete enough to use in our analyses. For the convenience sample, surveys were distributed by SESRC to all remaining certified organic operations over email, with postcard reminders. The survey was also advertised by Organic Seed Alliance (OSA) and OFRF. We received a total of 611 responses from the convenience sample, 230 of which included complete enough data for our analysis. In total we had 415 usable responses.

The breakdown between random and convenience sampling used in our analysis is summarized in Table S1. We consider the effect of sampling method, as well as survey distributor, through models in Section 2.

Table S1: Convenience and random samples, usable responses

Year	Total complete responses	Convenience responses (%)	Random responses (%)
2010	539	539 (100 %)	0 (0 %)
2015	771	649 (84 %)	122 (16 %)
2020	415	230 (55 %)	185 (45 %)

2. Checking for biases induced by data collection methods

In order to test how our data collection methods may bias the sample, we consider the effect of the sampling approach and survey distributor in our model. First we model the effect of sampling method with our 2015 and 2020 survey data (the only survey periods where we had both sampling approaches) by including a term ("Random data collection (vs. convenience)") for responses that were gathered by random sampling compared to convenience sampling (Table 2, Model 1). We see a small and insignificant effect for sampling method, and negligible changes in the results between the models run with and without the sampling consideration (Table 2, Model 2), suggesting no significant effect on results based on sampling approach.

Regarding concerns of survey distributor (i.e. that the distribution by Organic Seed Alliance (OSA) elicited responses from a select type of organic farmer), we model this effect with our 2010 and 2015 survey data (the only survey periods that identified specific distributors) by including a term ("Distributor: OSA (vs. Other)") for responses that were gathered through OSA channels compared to all others (Table 2, Model 3). We see a small and insignificant effect for distributor, and negligible changes in the results between the models run with and without the distributor consideration (Table 2, Model 4), suggesting no significant effect on results based on survey distributor.

Table S2: Beta regression model results including sampling and distributor effects

	Proportion of organic land planted with organic seed			
	Including sampling method (2015, 2020)	Without sampling method (2015, 2020)	Including survey distributor (2010, 2015)	Without survey distributor (2010, 2015)
	(1)	(2)	(3)	(4)
Ability	0.685***	0.687***	0.516***	0.506***
	(0.146)	(0.146)	(0.149)	(0.149)
Duty	0.249***	0.248***	0.238***	0.236***
	(0.048)	(0.047)	(0.043)	(0.043)
Ability x Duty	−0.121***	−0.122***	−0.094**	−0.091**
	(0.041)	(0.041)	(0.042)	(0.042)
Certifier request	−0.012	−0.011	−0.137*	−0.134*
	(0.079)	(0.079)	(0.074)	(0.074)
Farm size (log acres)	−0.128***	−0.128***	−0.130***	−0.128***
	(0.024)	(0.024)	(0.023)	(0.023)
Crop diversity	0.118*	0.118*	0.135**	0.136**
	(0.069)	(0.069)	(0.063)	(0.063)
Seed saved or traded (%)	0.057	0.057	0.091***	0.090**
	(0.039)	(0.039)	(0.035)	(0.035)
Years certified organic	0.016	0.016	0.038	0.038
	(0.036)	(0.036)	(0.038)	(0.038)
Forage crops (vs. Field)	−0.962***	−0.960***	−1.038***	−1.042***
	(0.117)	(0.116)	(0.108)	(0.108)
Vegetable crops (vs. Field)	−0.579***	−0.580***	−0.624***	−0.634***
	(0.125)	(0.125)	(0.119)	(0.119)
Western region (vs. North Central)	−0.279***	−0.278***	−0.342***	−0.383***
	(0.103)	(0.103)	(0.103)	(0.099)
Northeastern region (vs. North Central)	−0.288***	−0.284***	−0.214*	−0.265**
	(0.108)	(0.107)	(0.120)	(0.115)
Southern region (vs. North Central)	−0.033	−0.033	−0.194	−0.222*
	(0.140)	(0.140)	(0.134)	(0.132)
Year: 2020 (vs. 2015)	0.049	0.059		
	(0.084)	(0.080)		
Random data collection (vs. convenience)	0.033			
	(0.091)			
Year: 2010 (vs. 2015)			−0.268***	−0.262***
			(0.080)	(0.080)
Distributor: OSA (vs. Other)			−0.122	
			(0.082)	
Constant	0.176	0.184	0.386**	0.366**
	(0.183)	(0.182)	(0.168)	(0.167)
Observations	1,186	1,186	1,310	1,310
R^2	0.245	0.245	0.249	0.247
Log Likelihood	2,071.773	2,071.708	2,029.437	2,028.338

Note: *p<0.1; **p<0.05; ***p<0.01

3. Sample representativeness

While the mix of sampling methods creates a challenge for determining the sample's representativeness of the population, we aim to demonstrate that our sample is representative by comparing the geographic distribution of operations (Table S3) and farm size distribution (Figure S1) between our sample and USDA Census data.

We first use USDA Census data of organic farmers from 2011, 2016, and 2021 to align with our 2010, 2015, and 2020 survey periods. These data are made available by the National Agricultural Statistics Service Quick Stats API. Across the three survey periods, the number of respondents represented approximately 6.3%, 5.9%, and 2.5% of the total number of organic farmers identified in the USDA Census in the related years. In Table S3 we compare the number of operations in our sample (n) to the population (N) by year and region. For each census period we calculated the standard deviation of the percentage of operations represented, which we used to identify regions that are over or under-represented. Over-represented year-region combinations are signaled by italic text formatting, while under-represented year-region combinations are signaled by bold text formatting. Only in 2010 do we see significant variation in representation across regions, with the North Central and Southern regions over-represented and the Northeast under-represented.

Table S3: Sample representativeness compared to USDA Censuses (2011, 2016, 2021)

Year	Region	Operations in sample (n)	Operations in USDA census (N)	Percent of operations represented (n/N)
2010	North Central	305	2986	*10.2*
2010	Northeast	51	1897	**2.7**
2010	South	51	540	*9.4*
2010	West	139	3296	4.2
2015	North Central	386	4689	8.2
2015	Northeast	141	3246	4.3
2015	South	75	1096	6.8
2015	West	189	4529	4.2
2020	North Central	174	5824	3
2020	Northeast	98	3888	2.5
2020	South	38	1209	3.1
2020	West	98	5273	1.9

Note:
Italic formatting sigals over-representation and bold formatting signals under-representation

Next, to understand farm size representativeness, we look at USDA Census organic tabulations, reported for as the 2012, 2017, and 2022 census tabulations.

In Figure S1 we compare these USDA Census data against our survey sample. These comparisons show a consistent over-representation of the smallest farms (1-9 acres), especially in the North Central and South, and under-representation of the largest farms. Still, our sample captures a distribution of organic farm sizes that includes medium and large farms, allowing us to understand the effect of farm size in our analysis. Additionally, an important consideration is that we calculate farm size as reported acreage planted to a grower's main crop, focusing on three major crop categories (field, forage, and vegetable). However, this approach likely underestimates acreage compared to USDA Census data, which reports all certified land. Furthermore these data are not crop specific and so USDA data also includes non-seed crop operations (i.e. orchards).

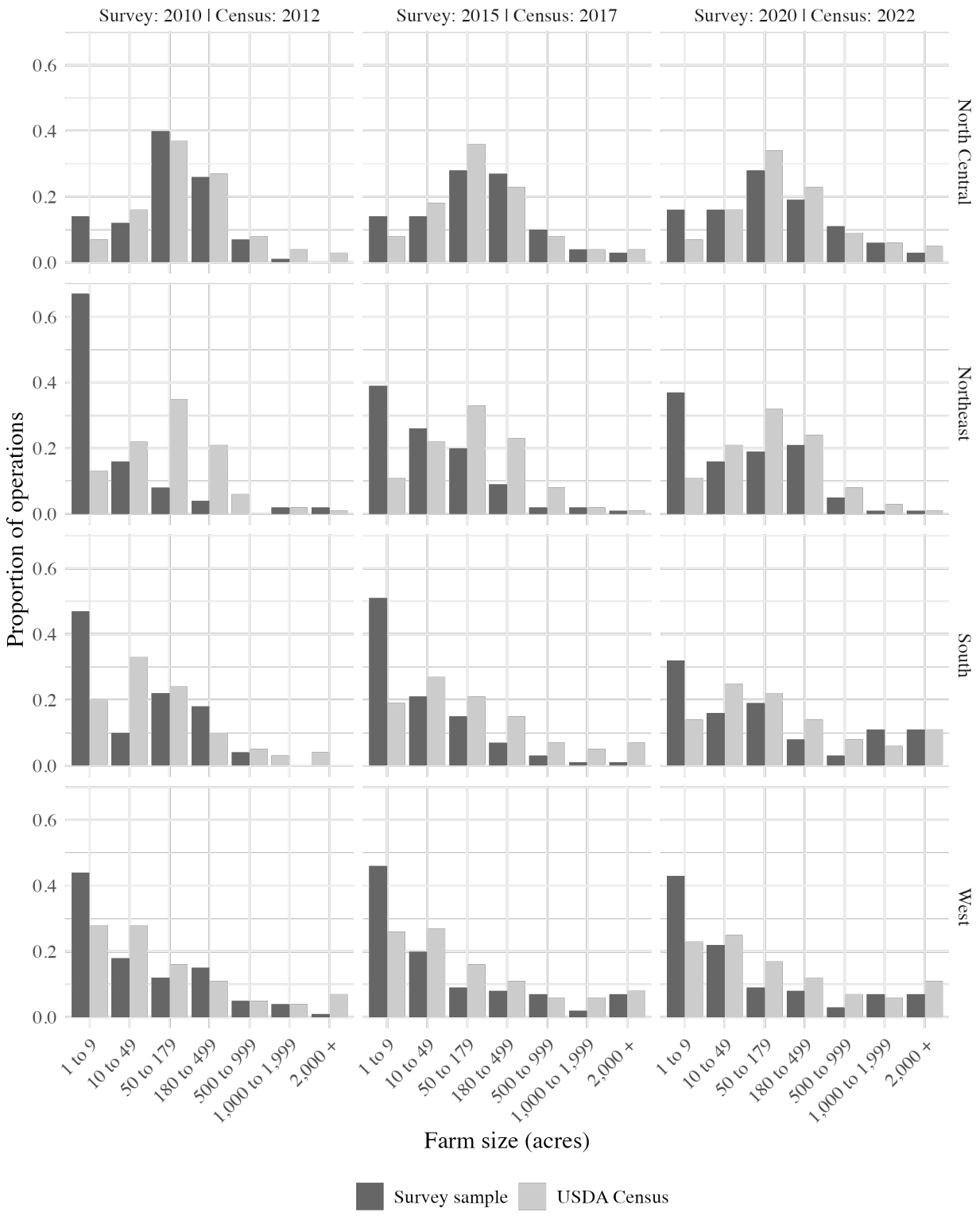


Figure S1: Sample farm size distribution compared to USDA Census data

Survey variation

4. Survey questions for key variables across survey periods

Acreage planted to organic seed

2010:
Q3a. How many organic acres in a given season do you have in annual vegetables and fruits? (Select one)
Q3b. What approximate percentage of the total acreage of the above crops is planted with certified organic seed (Select one)?
___ 0 ___ 1-25% ___ 25-50% ___ 51-75% ___ 76-100%

2015:
Q3. Last year, did you have any organic acres in annual vegetables? (Y/N)
The following two questions are conditional on the previous question:
Q4. How many organic acres did you have in annual vegetables last year? (Open response)
Q5. Last year, what approximate percentage of total acreage of annual vegetables was planted with certified organic seed?

2020:
Q5. How many certified organic acres did you have in annual vegetable crops last year? (Open response)
The following question is conditional on the previous question:
Q4A. Last year, approximately what percentage of your certified organic annual vegetable crops were planted with certified organic seed?

Duty

2010:
Q17. In your opinion, to what degree is using organic seed in organic systems important to the integrity of organic food product?
___ Very important/high priority ___ Important/medium priority ___ Not very important/low priority ___ Not at all important/not a priority

Q18. In your opinion, are varieties bred for organic system management important to the overall success of organic agriculture?
___ Very important/high priority ___ Important/medium priority ___ Not very important/low priority ___ Not at all important/not a priority

2015 / 2020:
Q27. / Q30. Please indicate the extent to which you disagree or agree with the following statements. (strongly disagree, disagree, neutral, agree, strongly agree)
- Organic seed is important to the integrity of organic food production
- Varieties bred for organic production are important to the overall success of organic agriculture

Ability

2010:
Q10. Using a scale of 1 = "not a factor" to 5 "significant factor" please share the reasons you have not purchased organic seed in the past. Please circle on response per reason below. (1 = not a factor; 2 = slight factor; 3 = moderate factor; 4 = more than moderate; 5 = significant factor)

2015:
Q23. For each reason listed below, how much of a reason was it for you to *not purchase* organic seed over the last three years (2011-2013)? (not a reason, slight reason, moderate reason, significant reason)

2020:
Q29. Over the last three years (2017-2019), how much were each of the following a factor in your decision NOT to purchase organic seed? (not a factor, slight factor, moderate factor, significant factor)

Explain how data were combined: * Acreage planted to organic seed: Categorical values in 2010 were... * Duty: 4-level categories from 1010 were converted into 5... * Ability: No cleaning necessary

5. Exploring variation in acreage planted to organic seed survey question

The survey question related to *acreage planted to organic seed*, our key dependent variable, is asked with different levels of precision across years. This change in precision requires further investigation, particularly to determine how respondents with split operations (i.e. farms in which part of their acreage is certified organic and another part is not) understand and respond to the question.

The question was intended (and interpreted in the paper) as it was posed in 2020: *Last year, approximately what percentage of your certified organic [vegetable / field / forage] crops were planted with certified organic seed?*. However, previous years omit the specific pointer to **certified organic** crops, and instead ask *Last year, what approximate percentage of total acreage of [vegetable / field / forage] was planted with certified organic seed?* (2015) or *What approximate percentage of the total acreage of the above crops is planted with certified organic seed?* (2010). These questions follow / are conditional on the previous question which is specifically about the organic acreage they have under production in a given crop. We expect that respondents will take the cue from the previous question and therefore respond about their organic seed use based only on their organic acreage.

To understand whether this question was interpreted as we expected, we carry out two explorations of the data: 1) Disaggregating results by year, with a focus on 2020 and 2) Identifying split operations with responses across multiple surveys.

Models disaggregated by year

The survey question related to acreage planted to organic seed is most precise in 2020, and therefore we would expect that the 2020 results should represent our overall expectations from the model. And indeed in Table S4 we observe results in 2020 that are consistent with the overall model results, suggesting that the questions are interpreted as we expect. Further, we can include a dummy variable for split operations in the 2020 data. We see that growers from split operations are less likely to plant their organic acreage to organic seed, we importantly observe that that the effects of our key variables of interest do not change.

Split operations with multi-year responses

To further explore the consequence of question imprecision on split operations, we tried to find respondents who we knew had split operations (based on 2020 response) and who also responded to previous surveys. However, because the surveys were not intended to be identifiable we had only limited methods for tracking repeated respondents (i.e. email addresses provided for follow-up contact, verified by location and crop matches). We found only one identifiable, repeat respondent who is known to have a split operation. We compared their answers between the 2015 and 2020 survey to determine whether they interpreted the questions differently.

In 2020, when asked, *Last year, approximately what percentage of your certified organic [vegetable / field / forage] crops were planted with certified organic seed?*, the respondent replied 100% for both crop types that they planted: field crops and forage crops. In 2015, when asked, *Last year, what approximate percentage of total acreage of [vegetable / field / forage] crop was planted with certified organic seed?*, the respondent also replied 100% for both crop types that they planted: field crops and forage crops. The consistent response across surveys, despite less precise question wording in 2015, suggests that this respondent took cues from the previous questions and understand the acreage in focus to be organic acreage.

Table S4: Model results by year

	Proportion of production acreage planted with organic seed			
	2010	2015	2020	2020 + split
Ability	0.138**	0.233***	0.389***	0.397***
	(0.056)	(0.048)	(0.064)	(0.063)
Duty	0.196***	0.219***	0.147**	0.144**
	(0.057)	(0.047)	(0.065)	(0.065)
Ability x Duty	−0.046	−0.095**	−0.114*	−0.120*
	(0.052)	(0.045)	(0.062)	(0.062)
Certifier request	−0.248**	−0.067	0.083	0.098
	(0.116)	(0.096)	(0.135)	(0.135)
Farm size (log acres)	−0.123***	−0.140***	−0.123***	−0.110**
	(0.039)	(0.029)	(0.045)	(0.045)
Crop diversity	0.169***	0.020	0.169**	0.162**
	(0.061)	(0.050)	(0.070)	(0.070)
Seed saved or traded (%)	0.174***	0.032	0.067	0.053
	(0.058)	(0.047)	(0.064)	(0.064)
Years certified organic	0.080	0.007	0.027	0.004
	(0.061)	(0.046)	(0.067)	(0.067)
Forage crops (vs. Field)	−0.787***	−1.250***	−0.584***	−0.592***
	(0.160)	(0.145)	(0.195)	(0.195)
Vegetable crops (vs. Field)	−0.480**	−0.824***	−0.235	−0.241
	(0.189)	(0.154)	(0.218)	(0.217)
Western region (vs. North Central)	−0.612***	−0.199	−0.385**	−0.362**
	(0.154)	(0.128)	(0.171)	(0.170)
Northeastern region (vs. North Central)	−0.180	−0.258*	−0.307*	−0.298*
	(0.217)	(0.137)	(0.169)	(0.169)
Southern region (vs. North Central)	−0.454**	−0.042	0.008	0.010
	(0.203)	(0.172)	(0.238)	(0.238)
Split operation				−0.412**
				(0.203)
Constant	0.903***	1.301***	0.888***	0.934***
	(0.125)	(0.102)	(0.142)	(0.143)
Observations	539	771	415	415
R^2	0.192	0.283	0.226	0.236
Log Likelihood	540.958	1,224.764	583.651	585.822

Note: *p<0.1; **p<0.05; ***p<0.01

Model checks

4. Correlations of all ability factors (Table S5)

Table S5: Correlations among organic seed sourcing ability factors

Barrier	Availability	Quantity	Quality	Form	Price
Quantity	0.50				
Quality	0.64	0.42			
Form	0.20	0.22	0.23		
Price	0.36	0.31	0.39	0.34	
Buyer requirement	0.41	0.38	0.27	0.21	0.22

5. Dissagregating allowable and non-allowable factors (Table S6)

Table S6: Beta regression model results disaggregating allowable and non-allowable ability factors

	Proportion of production acreage planted with organic seed			
Ability: Allowable	0.212***	0.408***	0.211***	0.310**
	(0.033)	(0.129)	(0.033)	(0.139)
Ability: Non-allowable	0.140***	0.137***	0.410***	0.371***
	(0.034)	(0.034)	(0.122)	(0.132)
Duty	0.198***	0.199***	0.186***	0.188***
	(0.039)	(0.039)	(0.039)	(0.039)
Ability: Allowable x Duty		−0.057		−0.029
		(0.036)		(0.039)
Ability: Non-allowable x Duty			−0.082**	−0.071*
			(0.036)	(0.039)
Certifier request	−0.043	−0.045	−0.046	−0.047
	(0.066)	(0.066)	(0.066)	(0.066)
Farm size (log acres)	−0.119***	−0.117***	−0.118***	−0.117***
	(0.021)	(0.021)	(0.021)	(0.021)
Crop diversity	0.181***	0.183***	0.183***	0.184***
	(0.056)	(0.056)	(0.056)	(0.056)
Seed saved or traded (%)	0.104***	0.104***	0.101***	0.101***
	(0.032)	(0.032)	(0.032)	(0.032)
Years certified organic	0.036	0.036	0.038	0.038
	(0.032)	(0.032)	(0.032)	(0.032)
Forage crops (vs. Field)	−0.918***	−0.920***	−0.919***	−0.919***
	(0.095)	(0.095)	(0.095)	(0.095)
Vegetable crops (vs. Field)	−0.510***	−0.499***	−0.493***	−0.489***
	(0.104)	(0.104)	(0.104)	(0.104)
Western region (vs. North Central)	−0.379***	−0.378***	−0.385***	−0.384***
	(0.086)	(0.086)	(0.086)	(0.086)
Northeastern region (vs. North Central)	−0.295***	−0.295***	−0.301***	−0.300***
	(0.095)	(0.095)	(0.095)	(0.095)
Southern region (vs. North Central)	−0.174	−0.171	−0.179	−0.177
	(0.116)	(0.116)	(0.116)	(0.116)
Year: 2010 (vs. 2015)	−0.217***	−0.214***	−0.219***	−0.217***
	(0.079)	(0.078)	(0.078)	(0.078)
Year: 2020 (vs. 2015)	0.091	0.089	0.089	0.088
	(0.080)	(0.080)	(0.080)	(0.080)
Constant	0.307**	0.304**	0.360**	0.351**
	(0.150)	(0.150)	(0.152)	(0.152)
Observations	1,725	1,725	1,725	1,725
R^2	0.236	0.237	0.238	0.238
Log Likelihood	2,934.334	2,935.591	2,936.934	2,937.200

Note: *p<0.1; **p<0.05; ***p<0.01

6. Dissagregating all barriers (Table S7)

Table S7: Beta regression model results disaggregating all ability factors

	Proportion of production acreage planted with organic seed
Ability: Availability (Allowable)	0.179***
	(0.038)
Ability: Quality (Allowable)	0.114***
	(0.037)
Ability: Quantity (Allowable)	0.019
	(0.035)
Ability: Form (Allowable)	−0.066**
	(0.033)
Ability: Price (Non-allowable)	0.065*
	(0.034)
Ability: Buyer Requirement (Non-allowable)	0.121***
	(0.034)
Duty	0.183***
	(0.039)
Certifier request	−0.043
	(0.066)
Farm size (log acres)	−0.108***
	(0.021)
Crop diversity	0.174***
	(0.056)
Seed saved or traded (%)	0.102***
	(0.032)
Years certified organic	0.040
	(0.032)
Forage crops (vs. Field)	−0.908***
	(0.094)
Vegetable crops (vs. Field)	−0.485***
	(0.105)
Western region (vs. North Central)	−0.382***
	(0.087)
Northeastern region (vs. North Central)	−0.320***
	(0.095)
Southern region (vs. North Central)	−0.159
	(0.116)
Year: 2010 (vs. 2015)	−0.208***
	(0.079)
Year: 2020 (vs. 2015)	0.107
	(0.080)
Constant	0.345**
	(0.151)
Observations	1,725
R^2	0.247
Log Likelihood	2,944.363
Note:	*p<0.1; **p<0.05; ***p<0.01